

MACHINE LEARNING



Introduction to Machine Learning

From Data to Deployment — Pipelines, Real-World
Applications, Challenges & Careers



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What Is Artificial Intelligence?

“What comes to mind when you hear Artificial Intelligence?”

- Robots
- ChatGPT
- Self-driving cars
- Siri
- Face recognition
- Video games
- Medical diagnosis

AI is a branch of computer science focused on building systems that perform tasks which typically require human intelligence.

These tasks include:

Learning

Reasoning

Problem-solving

Understanding
language

Recognizing images

Making decisions

Key idea: AI does not “think” like humans — it learns patterns from data and uses them to predict or decide.

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Human Intelligence vs. AI

HUMAN	AI
Learns from experience	Learns from data
Sees objects	Detects objects in images
Understands speech	Speech recognition
Makes decisions	Predicts or recommends
Drives a car	Autonomous driving



AI doesn't “think” like humans.

It learns patterns from data and uses those patterns to make predictions or decisions — no lived experience required.

Real-Life Examples of AI



Recognize a face

Phone Face Unlock



**Understand
speech**

Siri, Google Assistant



**Translate
languages**

Google Translate



**Recommend
movies**

Netflix



Detect spam

Gmail



Generate images

DALL·E



Write text

ChatGPT

AI Is Everywhere

“How many times do you think you use AI every day?”



Morning

Face Unlock recognizes your face



Traveling

Google Maps predicts traffic & routes



Social Media

YouTube suggests videos you'll like



Shopping

Amazon recommends products



Email

Spam filter sorts your inbox



Banking

Fraud detection flags suspicious activity



Entertainment

Spotify builds personalized playlists



Chatbots

ChatGPT answers, writes, summarizes

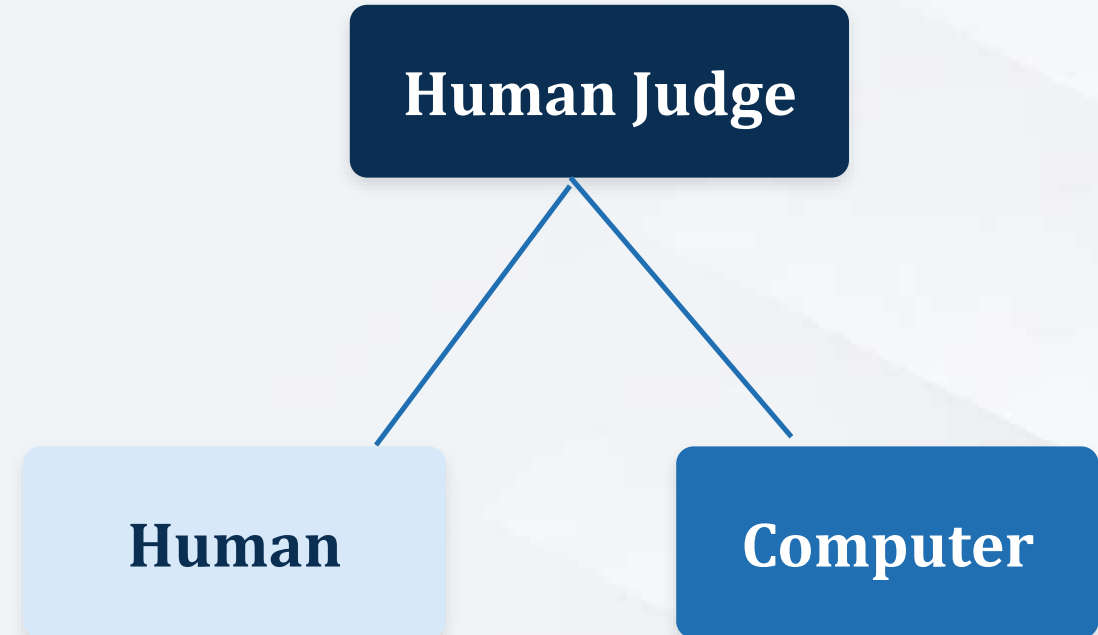
1950 — The Beginning

ALAN TURING

“Can a machine think?”

This was exactly Alan Turing’s question. He proposed what is now known as the Turing Test — a way to judge whether a machine can exhibit intelligent behavior indistinguishable from a human.

Imagine chatting online. One participant is a human, the other a computer. If the judge can’t tell which is which — the computer passes the test.



Can the judge tell the difference?

From a Named Field to Generative AI

1956

1980s

1997

2012

2016

2022

The field is officially named “Artificial Intelligence.” Researchers believed machines could become intelligent within decades — it turned out far harder.

Expert Systems follow thousands of hand-written IF-THEN rules (e.g. IF fever AND cough THEN maybe flu). No learning — every rule written manually.

IBM Deep Blue beats world chess champion Garry Kasparov. Computers prove they can outperform humans at specific tasks.

The Deep Learning Revolution: AlexNet wins the ImageNet competition, dramatically improving image recognition and sparking modern AI.

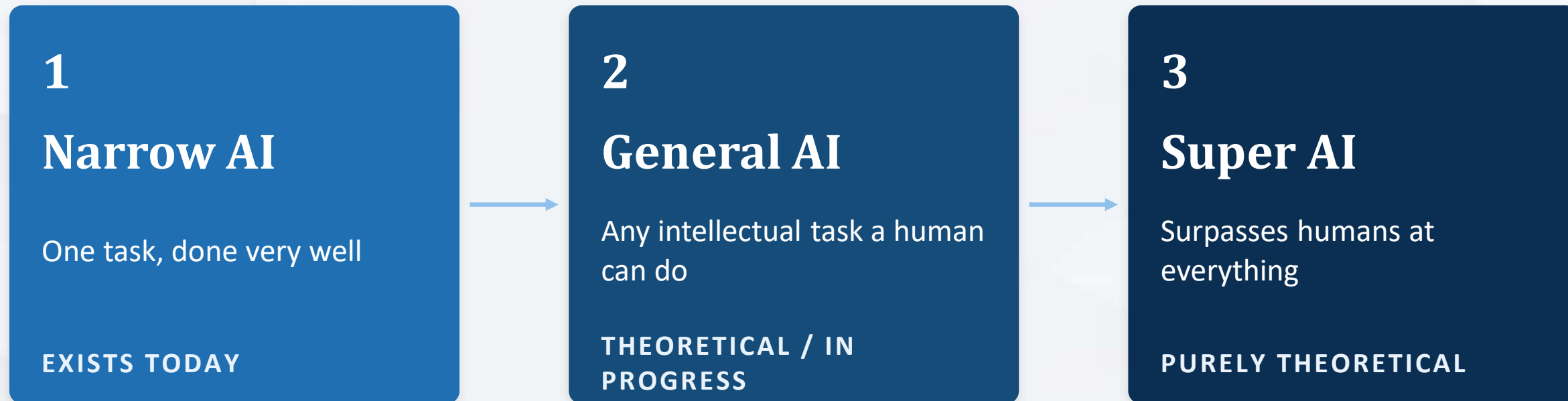
AlphaGo defeats Lee Sedol at Go — a game many experts believed would remain impossible for AI.

ChatGPT launches. Millions of people use AI directly for the first time — AI becomes available to everyone.

Today: Generative AI can produce text, images, music, video, and code.

Not All AI Is Equally Intelligent

Scientists usually divide AI into three levels of capability:



Only Narrow AI exists today. General AI and Super AI remain research goals and thought experiments.

Narrow AI (Weak AI)

THE ONLY TYPE THAT EXISTS TODAY

AI designed to perform one specific task very well.

Even ChatGPT — despite doing many language tasks — is still Narrow AI: it is specialized and lacks human-level general intelligence across all domains.

EXAMPLES

ChatGPT	Siri	Face Recognition	Google Translate
Chess AI	Recommendation systems	Spam detection	

CHARACTERISTICS

- ✓ Excellent at one area
- ✗ Cannot truly understand
- ✗ Cannot think like humans
- ✗ Cannot transfer knowledge to unrelated tasks

A chess AI can defeat world champions — but cannot drive a car. A self-driving car can navigate roads — but cannot diagnose diseases. Each AI specializes in its own domain.

General AI (AGI)

Artificial General Intelligence can perform any intellectual task that a human can — without retraining.

IMAGINE ASKING IT TO...

Solve math

Write poetry

Cook

Drive

Teach

Invent something

Play sports

Learn a new language

CHARACTERISTICS

- Learns any task
- Understands concepts
- Reasons logically
- Transfers knowledge
- Adapts to new situations

*AGI does not exist today.
Researchers are actively working
toward it, but there is no
consensus on when — or if — it
will be achieved.*

Super AI

An intelligence that surpasses humans in every intellectual task — smarter than scientists, doctors, engineers, and artists combined.

CHARACTERISTICS

- Learns faster than any human
- Solves problems humans cannot
- Creates new scientific discoveries
- Thinks beyond human capabilities

CURRENT STATUS

Theoretical.

It has never been created. Most discussions about Super AI come from research, philosophy, and science fiction — not deployed systems.

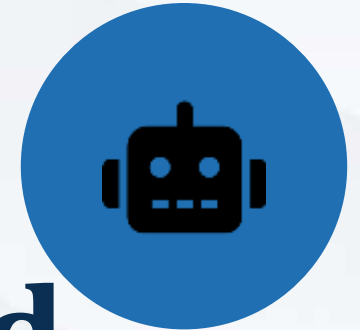
Comparing the Three Levels

FEATURE	NARROW AI	GENERAL AI	SUPER AI
Exists today?	✓ Yes	X No	X No
Performs one task	✓	X	X
Learns many tasks	X	✓	✓
Human-level intelligence	X	✓	Beyond humans
Example	ChatGPT, Siri	None	None

Classify Each Example

Show these examples and ask students to classify each one:

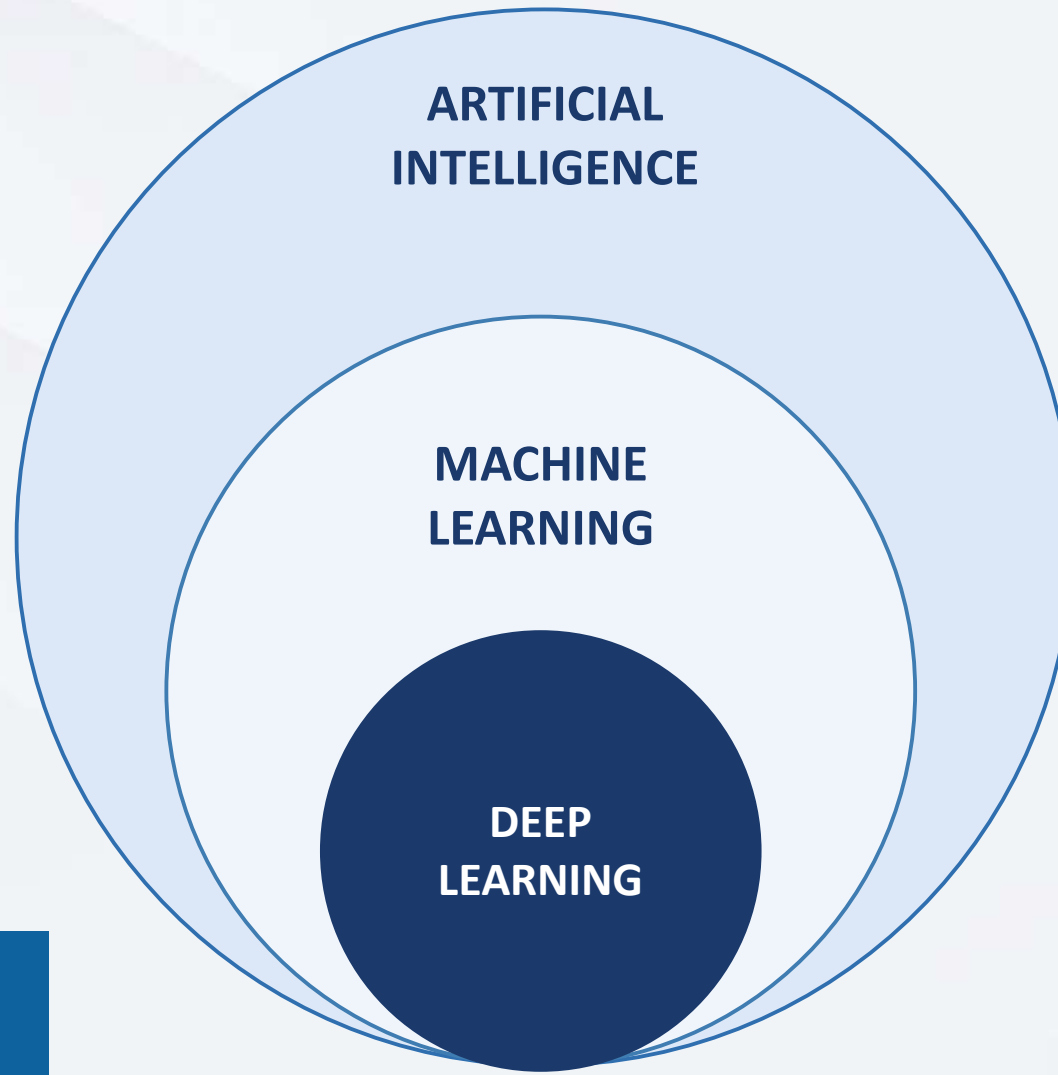
EXAMPLE	TYPE
ChatGPT	Narrow AI
Netflix Recommendation	Narrow AI
Face Recognition	Narrow AI
A robot that can learn any job like a human	General AI
An AI that solves problems no human can solve	Super AI



Are AI, Machine Learning, and Deep Learning the same thing?

Three Nested Fields

Each inner circle is a subset of the outer one



AI

Any technique that makes a computer mimic human intelligence — doesn't always learn.

Machine Learning

A subset of AI that learns patterns from data instead of following hand-written rules.

Deep Learning

A subset of ML using neural networks — powerful for images, speech, and language.

Artificial Intelligence: The Big Umbrella

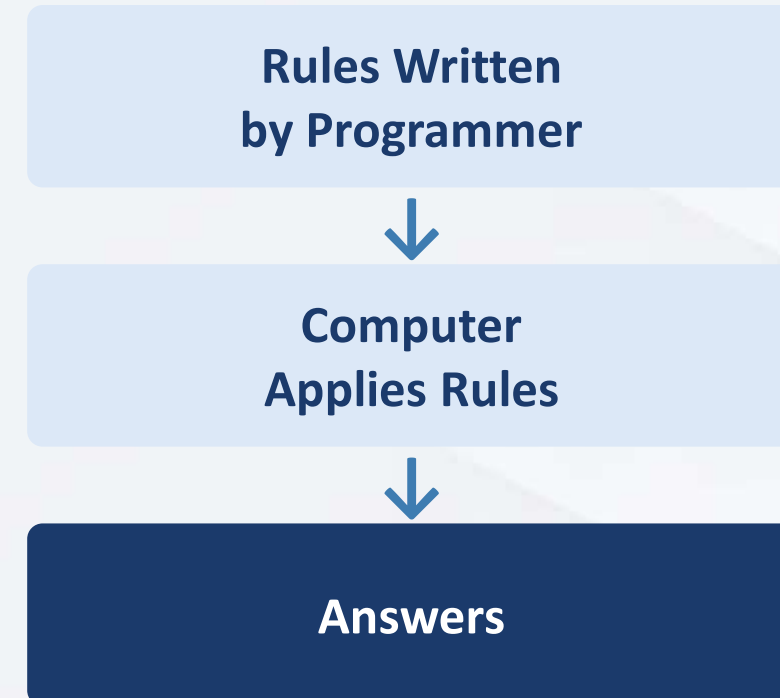
AI is any technique that enables a computer to mimic human intelligence — and it doesn't always learn.



Example: Chess Program

- A programmer writes thousands of chess rules
- The computer can then play chess
- It is AI — but not Machine Learning, because it never learns

Rule-Based Flow



Machine Learning: Learning from Examples

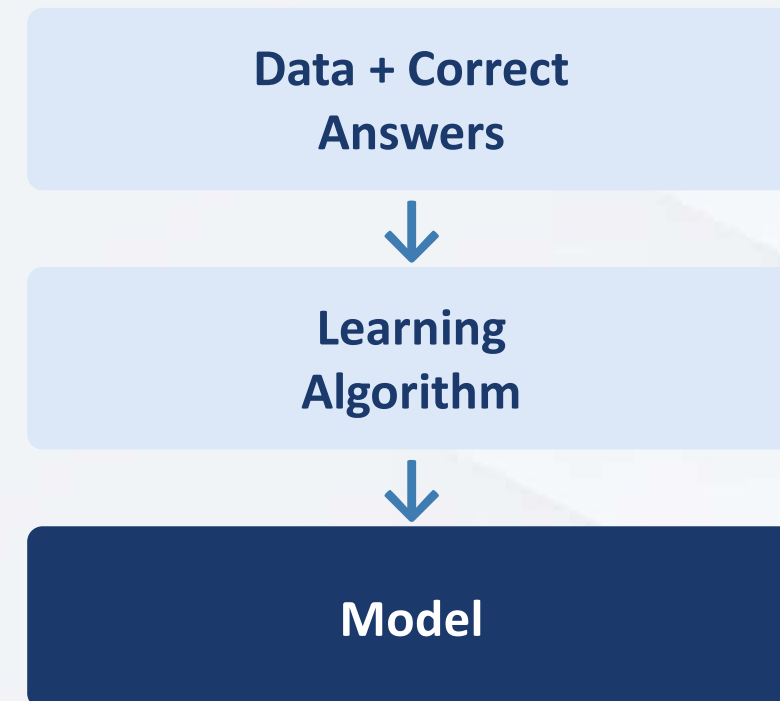
Instead of writing rules, we give the computer examples — it discovers the rules itself.



Example: Spam Detection

- Instead of writing: IF contains “FREE” THEN spam
- We collect 50,000 emails, some labeled Spam / Not Spam
- The computer studies them and learns the patterns automatically

Machine Learning Flow



“Machine Learning learns from experience (data).”

Deep Learning: Neural Networks at Scale

A subset of ML that uses Artificial Neural Networks, inspired by the human brain — especially powerful for:



Images



Speech



Videos



**Natural
Language**



**Generative
AI**

Real-world examples:

- ChatGPT
- Image generation
- Face recognition
- Voice assistants

A Real-Life Analogy: Recognizing Cats



AI

- Follows human-defined rules.
- Makes decisions based on explicit logic.
- Does not learn from data on its own.



Machine Learning

- Learns from labeled examples.
- Relies on human-designed features.
- Works well with moderate-sized datasets.



Deep Learning

- Learns directly from raw data.
- Automatically discovers features.
- Usually requires much larger datasets and more computing power.

AI vs ML vs Deep Learning at a Glance

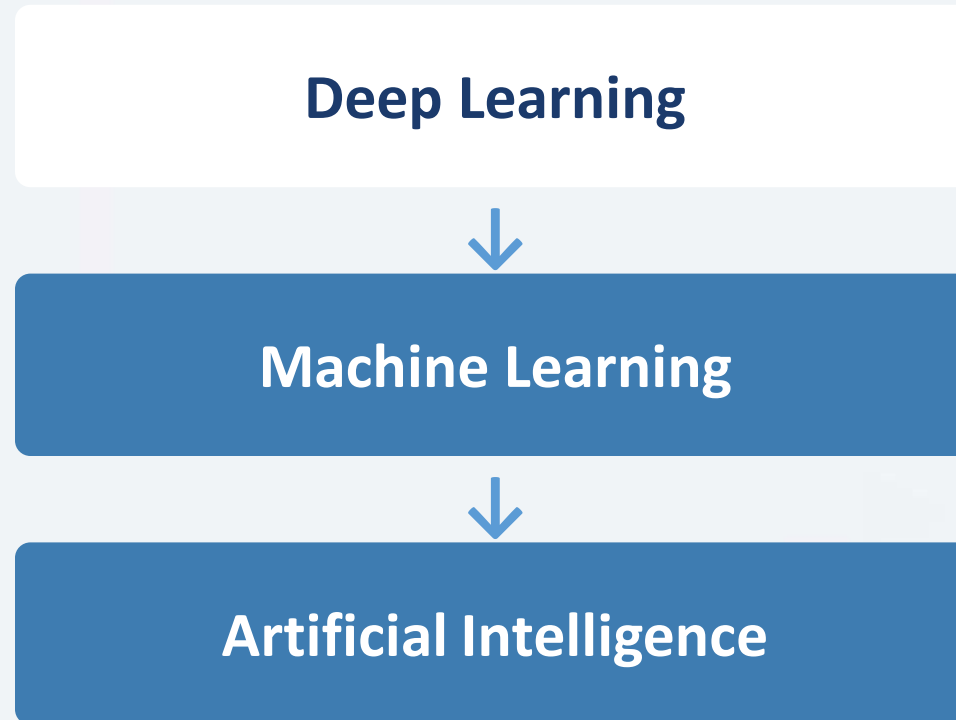
Artificial Intelligence	Machine Learning	Deep Learning
Broad field	Part of AI	Part of ML
Can use rules	Learns from data	Learns using neural networks
Doesn't always learn	Learns automatically	Learns complex patterns
Oldest field	Newer	Newest

Note: the fields are nested — a technique in Deep Learning is also Machine Learning and AI.

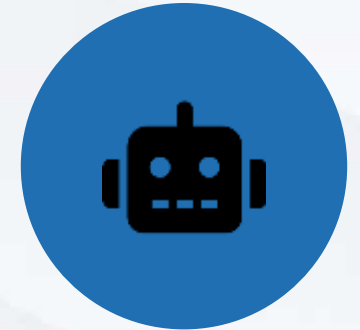
Which Category Does It Belong To?

Example	Category
Calculator	Not AI
Chess game using hard-coded rules	AI
Spam detector trained on emails	Machine Learning
Face recognition using CNN	Deep Learning
ChatGPT	Deep Learning (also ML and AI)

Which Category Does ChatGPT Belong To?



It belongs to all three.



What Is Machine Learning?

How Can We Teach a Computer to Recognize Cats?

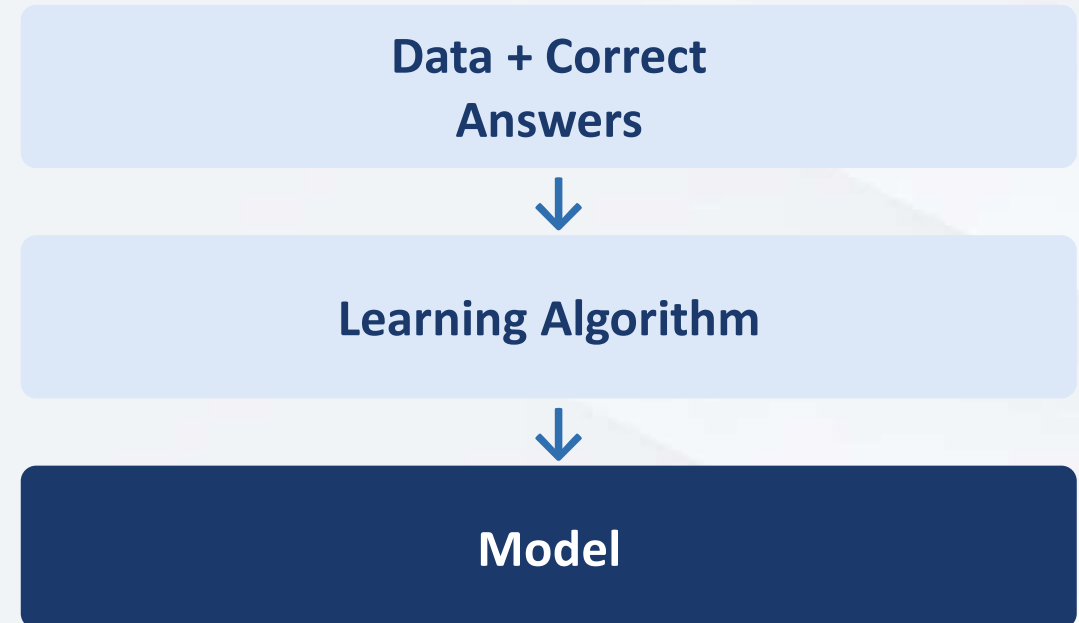
Most people answer: “write rules.” But what about kittens, black cats, sleeping cats, or toy cats? You'd need thousands of rules — impossible.

Traditional Programming



The programmer must write every single rule.

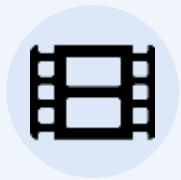
Machine Learning



The computer discovers the rules itself, then predicts on new images.

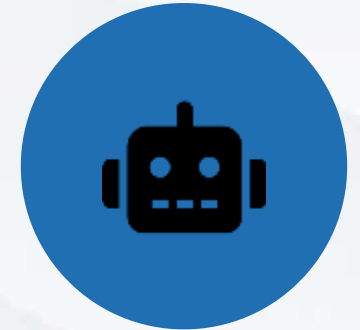
A Simple Definition

Machine Learning is the process of enabling computers to learn patterns from data and use those patterns to make predictions or decisions — without being explicitly programmed for every situation.



Real-Life Example: Netflix

Does Netflix know your favorite movie? No. It studies what you watch, what you skip, and how you rate things — then it predicts what you might like next. That's Machine Learning in action.



Types of Machine Learning?

A. Supervised Learning

Like teaching a child with flashcards — every example comes with the correct answer.

Input	Label
House features	House price
Email	Spam / Not Spam
Medical scan	Healthy / Diseased
Student data	Pass / Fail

Goal: predict the correct label for new, unseen data.



Algorithms We'll Study

- Linear Regression
- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

B. Unsupervised Learning

No labels, no answers — the computer discovers hidden patterns by itself.

Customer Segmentation Example

Customers



Similar Behavior



Groups

Other examples: grouping similar products, news topic clustering, finding unusual behavior, market basket analysis.



Algorithms

- K-Means
- DBSCAN
- Hierarchical Clustering
- PCA (dimensionality reduction)

Note: PCA is not a clustering algorithm — it reduces features while preserving as much information as possible.

C. Reinforcement Learning

Like teaching a dog: reward good behavior, withhold reward for mistakes — eventually it learns.



AlphaGo

Didn't memorize games — it learned through millions of games, receiving rewards for winning.



Other Examples

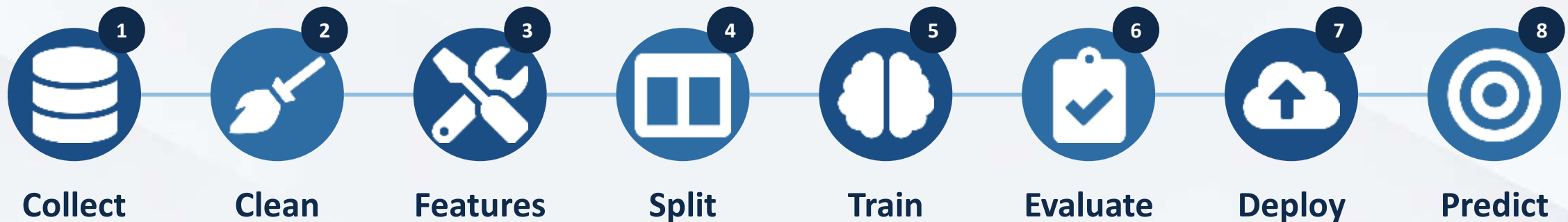
Chess · Go · Self-driving cars · Robots · Game AI

The Three Types, Side by Side

Supervised	Unsupervised	Reinforcement
Has labels	No labels	Rewards
Predicts outputs	Finds hidden patterns	Learns by trial and error
Teacher provides answers	No teacher	Feedback through rewards
Spam detection	Customer segmentation	Game playing

The Machine Learning Pipeline

“Building an ML model is like building a house — you can't start with the roof.”



Most of the real work happens before training even starts.

Data collection, cleaning, and feature engineering typically consume more time than fitting the model itself — and largely determine how good the final predictions can be.

Step 1-2: Collect & Clean Data



Collect Data

Data is the fuel of Machine Learning — a model cannot learn without it.

Common sources:

- Customer records
- Medical images
- House prices, weather data
- Sales history, sensor readings

Example — predicting house prices, collect:

Area · rooms · location · age · garage · previous selling price



Clean Data

Real-world data is messy — cleaning improves data quality.

Common problems:

- Missing values
- Duplicate records
- Incorrect values or outliers
- Inconsistent formats

Example — an “Age” column:

Age
20
?
18
200 ← likely an error

Step 3-4: Feature Engineering & Split Data



Feature Engineering

Features are the input variables a model learns from. Better features often help more than a fancier algorithm.

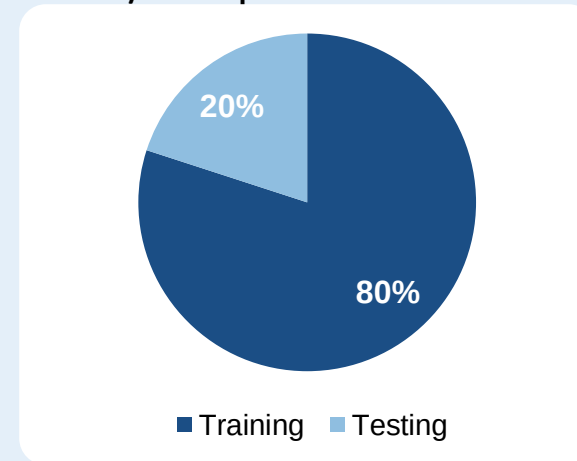
Example:

- Date of purchase → Day of week, Month, Weekend/weekday
- Birth Year = 2004 → Age = 22 (often more useful)



Split Data

Never train and test on the same data — usually an 80 / 20 split.



Analogy: a student studies from practice questions; the final exam must contain new questions — otherwise you can't tell if they actually learned.

Step 5-6: Train & Evaluate the Model



Train the Model

The algorithm studies the training data and learns patterns — no one explicitly programs every rule.

Example — spam detection. The model learns from:

- Certain words
- Email length
- Links
- Sender information



Evaluate the Model

“How do we know if the model is good?” We test it on unseen data.

Common evaluation metrics:

- Accuracy
- Precision & Recall
- F1-score
- Mean Squared Error (regression)

Step 7–8: Deploy & Predict



Deploy the Model

A trained model is useful only once people can actually use it — by integrating it into a real application:

- Website / mobile app
- Hospital system
- Banking software



Predict

The model receives new data and produces predictions:

New customer



Model



“Likely to buy” or “Not likely to buy”

Summary: the pipeline isn't just about training a model — most of the work is preparing the data and making the model usable.

AI Applications by Industry



Healthcare

Disease diagnosis,
medical imaging, drug
discovery



Finance

Fraud detection, credit
scoring, stock prediction



Education

Personalized learning,
automated grading,
tutoring



Agriculture

Crop monitoring, disease
detection, smart
irrigation



Manufacturing

Predictive maintenance,
quality inspection,
robotics



Transportation

Self-driving cars, traffic
prediction, routing



Retail

Recommendations,
demand forecasting,
inventory



Cybersecurity

Intrusion, malware &
phishing detection

Challenges of AI (1/2)



Bias

If training data is biased, the model learns and repeats those unfair patterns



Privacy

AI often relies on personal data — medical records, location, face and voice — raising concerns about collection and storage.



Security

Systems can be attacked: adversarial examples that fool classifiers, data poisoning, and model theft.



Fake Images & Video

Generative AI can create convincing deepfakes, spreading misinformation and blurring real vs. fake.

Challenges of AI (2/2)



Hallucinations

LLMs sometimes generate incorrect or fabricated information while sounding confident. Always verify AI output for important decisions.



Ethics

Should AI make medical decisions? Who is responsible if an autonomous car causes an accident? How do we ensure fairness and transparency?



Job Automation

AI automates repetitive tasks while creating new roles requiring AI skills — it is more likely to change jobs than replace all workers.



Explainability

Models can be accurate yet hard to explain — critical in fields like healthcare and finance where decisions affect people's lives.

AI Myths — True or False?



AI will replace everyone.



AI automates some tasks and transforms many jobs, while creating new opportunities.



AI always knows the truth.



AI can make mistakes and generate inaccurate information.



Bigger models are always better.



The best model depends on the problem, data, speed, cost, and accuracy needs.



AI understands exactly like humans.



Today's AI learns statistical patterns — it has no human consciousness or understanding.



AI is only for programmers.



Doctors, engineers, marketers, teachers, and designers all use AI tools today.

Careers in AI (1/2)



Data Analyst

Analyze data, build dashboards, generate business insights.

*Excel · SQL · Python ·
Power BI · Tableau*



Data Scientist

Explore data, build predictive models, solve business problems.

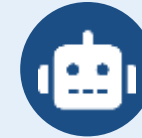
*Statistics · Python ·
Machine Learning*



ML Engineer

Build, optimize, and deploy ML models with software engineers.

*Model deployment ·
Software integration*



AI Engineer

Develop AI-powered apps combining models with software systems.

*Chatbots ·
Recommenders ·
Assistants*

Careers in AI (2/2)



NLP Engineer

Language technologies — chatbots, translation, text summarization, sentiment analysis.



Computer Vision Engineer

Image & video understanding — face recognition, medical imaging, autonomous vehicles.



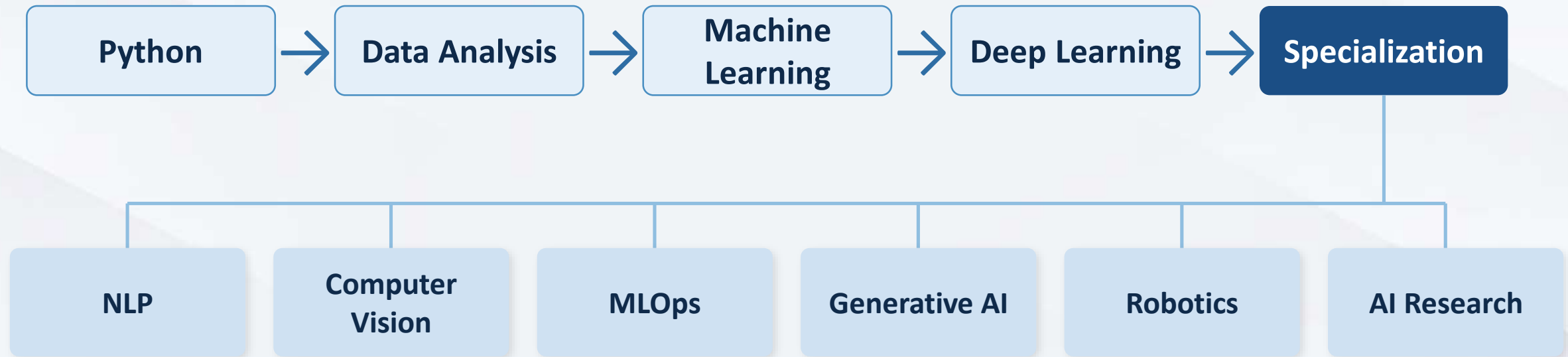
MLOps Engineer

Deploy, monitor, and maintain ML models in production — the bridge between ML and software ops.



Research Scientist

Invent new AI algorithms and publish research. Usually requires an MSc or PhD.



A typical path builds general skills first — Python, data analysis, ML foundations — before branching into a specialization aligned with your interests.

Thank you